

AC-OS v0.2 — Aspectual Counting Operating System (Research-Grade Package)

Executive Summary

AC-OS turns identity/count disputes into a **constraint-selected bridge problem** with principled certificates. We model aspects as functors to finite sets, bridges as **lax natural relations** (modalities: *id*, *map*, *evidence*), and strictness as **cannot-link** relations. Selection becomes core-guided Weighted Partial MaxSAT / ILP over a **conflict hypergraph**; explanation is via **unsat cores/MUS**, **k-shadow** (arity-bounded relaxations), **dominance** and **unavoidable-tail** certificates. v0.2 adds explicit situation scope, modality constraints, simplified certificates, and context-merge provenance.

1) Novelty + Practicality

Novel (as a package)

- **Counting = audited bridge selection.** “No bridge, no transfer.” Paradoxes reduce to selecting a maximum-weight feasible subset with proofs.
- **Interaction order as a diagnostic.** Use minimal conflict **arity** and **k-shadow** convergence to certify when pairwise governance fails.

Practical

- Ship v1 with Weighted MaxSAT or ILP, MUS/core extraction, and human-readable witness paths.
 - k-shadow gives **monotone tightening**: fast approximate decisions with explicit convergence tests.
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2) Enhanced Version (v0.2 highlights)

- **Situation scope:** bridge `scope` $\in$ {per-situation, cross-situation, global}; per-edge `situation` required in the first case.
 - **Modality constraints:** `map` requires `max_out_degree`; `evidence` requires `evidence_threshold` (bridge-level), with optional per-situation thresholds.
 - **Certificate simplification:** `k_shadow` reports {`k`, `objective`}; cores have stable string IDs; methods renamed to neutral (`fractional_matching`, `lp_dual`, `hitting_set`).
 - **Context join semantics:** provenance via `ancestors`, `merge_strategy`, and semantic weight merges (`sum`, `left_prefer`, `right_prefer`, `max`).
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3) Internal Critique (stress points)

- **Closure overcommit:** not all bridges should induce equivalence. We address this with **modalities** and modality-specific closure rules.
 - **Partiality vs functoriality:** bridges modeled as **lax** natural relations to legitimate lossy/partial transfer.
 - **Arity terminology:** use *conflict arity* and *practical conflict arity* (k where k-shadow = optimum) to avoid overload with classical Helly.
 - **Dominance cost:** certify dominance **relative to a discovered core basis** (solver traces), not the full MUS family.
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4) Final Spec + Predictions

Semantics

- **Situations:** category $\mathcal{O}_c$ (context-scoped).
- **Aspects:** functors $U_\alpha : \mathcal{O}_c \rightarrow \mathbf{FinSet}$.
- **Bridges:** candidates $\tau^{(b)} : U_\alpha \Rightarrow U_\beta$, modality $\in \{\text{id}, \text{map}, \text{evidence}\}$, scope as above.
- **Strictness:** cannot-links $S_{\alpha, X}$.
- **Selection:** choose $\Gamma \subseteq B_c$ maximizing total weight subject to **no cannot-link is forced** by the modality closure under Γ .

Predictions

1. **Arity spikes under context joins:** benign contexts often have arity 2-3; joins trigger higher arity exactly when governance collides.
 2. **Concentrated unavoidable drops:** #-strict contexts produce a small, interpretable set of excluded bridges.
 3. **k-shadow as thermometer:** early convergence $\Rightarrow$ pairwise suffices; delayed convergence $\Rightarrow$ genuine higher-order coupling.
 4. **Explanation stability correlates with core stability** under small weight perturbations.
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5) Mathematical Overview (formal spine)

5.1 Closure & feasibility

Let $V_X = \bigsqcup_\alpha U_\alpha(X)$. Each bridge b contributes edges $E_{b,X} \subseteq V_X \times V_X$. Define modality-aware closure $R_{\Gamma, X} = \text{Cl}_{\text{mod}}(\bigcup_{b \in \Gamma} E_{b,X})$.
Feasible iff no $(u, v) \in S_X$ lies in $R_{\Gamma, X}$.

5.2 Conflict hypergraph

Minimal conflicts (MUS-analogues) $\mathcal{M}_c$ yield $H_c = (B_c, \mathcal{M}_c)$. Selection $\equiv$ **maximum-weight independent set** with constraints $\sum_{b \in M} x_b \leq |M| - 1$.

5.3 k-shadow

Enforce only cores with $|M| \leq k$. Objectives OPT_k decrease monotonically to OPT . The **practical conflict arity** is the smallest k where equality holds.

5.4 Certificates

- **Cores with witness paths** (shortest-path forcing a cannot-link).
- **Dominance (basis-relative)**.
- **Unavoidable tail** via fractional matchings / LP duals / hitting sets.

6) Fastest Validation Path

1. **Executable core**: data ingestion, modality closures, feasibility check.
2. **Solver**: Weighted Partial MaxSAT (preferred) or ILP; adopt **core-guided** flow.
3. **Artifacts**: cores with witness paths; k-shadow trajectory ($k=2,3,4,\dots$).
4. **Datasets**: (i) Synthetic Theseus (tunable arity); (ii) ER/record-linkage with must/cannot-links.
5. **Success**: optimum found; explanations stable; k-shadow distinguishes MUS2 vs higher-order.

7) Artifacts (v0.2)

- **Schemas**: `acos_instance.v0.2.schema.json`, `acos_certificate.v0.2.schema.json`
- **Utilities**: `acos_validate.py` (schema + semantic checks), `acos_witness.py` (shortest-path witness).
- **Examples**: `example_v0.2_instance.json` (with `map` + `evidence`), `example_v0.2_certificate.json`, plus `example_v0.2_core_based.wcnf` and `example_v0.2_from_cores.lp`.

Note: Downloads were shared in-chat; keep these filenames synchronized with your repo.

8) CLI Sketch

```
acos solve
--instance example_v0.2_instance.json
--model weighted_maxsat|ilp
--encoding core_based|full
--k-shadow 2..6
--emit witness
--out certificate.json
--validate
```

9) Witness Generation (algorithmic note)

1. Build mixed-mode graph from selected bridges (`id` undirected; `map` directed; expand `evidence` into active `id` edges once thresholds are met).
 2. For each cannot-link pair, run BFS/shortest-path; the first reachable pair yields a minimal witness path.
 3. Store `(violation_pair, path)` in the certificate core.
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10) Benchmarking & Prediction Tests

- **Arity spikes:** run k-shadow on two benign contexts and on their join; visualize OPT_k .
 - **Core stability under perturbations:** jitter weights; compute Jaccard of core memberships.
 - **Human trust vs artifacts:** A/B test explanations with/without witness paths and unavoidable-tail bounds.
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External References (selected)

- Core-guided / iterative MaxSAT surveys and solvers; MUS/MCS/MSS definitions; DRAT/DRAT-trim proof infrastructure; abstract argumentation; constrained ER; hypergraphs, hitting sets, and Helly-style properties; profunctors/spans in category theory.
- See the reference list at end; URLs are provided for reproducibility.**
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Reference List (with URLs)

- Morgado, Heras, Marques-Silva, Planes. *Iterative and Core-Guided MaxSAT Solving: A Survey and Assessment* (Constraints). PDF: https://sun.iwu.edu/~mliffito/publications/constraints_morgado_maxsat_survey.pdf
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- Dung. *On the Acceptability of Arguments...* (AI 1995). PDF: <https://cse-robotics.engr.tamu.edu/dshell/cs631/papers/dung95acceptability.pdf>
- Wang et al. *Provenance-Aware Entity Resolution* (DASFAA 2015). PDF: <https://users.cecs.anu.edu.au/~u5170295/papers/dasfaa-wang-2015.pdf>
(See also standard ER/linkage surveys.)

- Berge. *Hypergraphs: Combinatorics of Finite Sets* (Elsevier 1989). Archive: <https://archive.org/details/hypergraphscombi0000berge>
 - Gyárfás. *A Note on Hypergraphs with the Helly Property*. PDF: https://users.renyi.hu/~gyarfas/Cikkek/12_Gyarfas_ANoteOnHypergraphsWithTheHellyProperty.pdf
 - nLab. *Profunctor*. <https://ncatlab.org/nlab/show/profunctor>
(Background on profunctors/spans as categorical relations.)
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